

Attention Is All You Need

# Transformers Overview

Replace recurrence and convolution with attention mechanisms.  
Capture diverse relationships in parallel using scaled dot-product attention and multi-head attention.

# Transformer Architecture

Stacked encoder and decoder layers  
interleave sublayers.  
Positional encodings provide sequence  
order information.



# Scaled Dot-Product Attention

Computes how each element in a sequence should attend to every other element.  
Transforms input embeddings into queries, keys, and values spaces.  
Scores and weighted to produce context-aware representations.

# Multi-Head Attention

Uses  $h$  parallel heads with learned linear projections for Q, K, and V.  
Captures different types of relationships simultaneously.  
Concatenates outputs and linearly projects back to the model space.

# Positional Encoding

Gives each token a unique position signal added to its embedding before attention layers.

Handles permutation-invariance and sequence order awareness in transformers.

# Encoder & Decoder Stacks

Encoders have two sub-layers, decoders have three.  
Residual connections and layer normalization ensure stable training.

# Miscellaneous

Time complexity per layer per token is  $O(n^2d)$ .

Memory complexity is  $O(n^2)$ , limiting very long sequences without modifications.